

Mustard

Brassica juncea / Brassica campestris — A Key Rabi Oilseed of Pakistan

A Complete Cultivation Guide for

Punjab · Sindh · Khyber Pakhtunkhwa · Balochistan

Compiled for Farmer Guidance & AI Advisory Use

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1. Introduction

Mustard, known locally as sarson or raya, is one of Pakistan's traditional oilseed and fodder crops, grown mainly during the winter (rabi) season. It is valued both for its oil, widely used in cooking and pickling, and for its greens, which are a popular winter vegetable dish (sarson ka saag) in Punjab. Mustard is a hardy crop that tolerates cooler temperatures, moderate drought, and somewhat saline soils better than many other oilseeds, which makes it a useful crop for marginal lands in Punjab, Sindh, and parts of Khyber Pakhtunkhwa. This guide explains how mustard is grown across Pakistan and how farmers can improve yield and oil quality.

2. Scientific Name

The mustard crops grown in Pakistan mainly belong to two species: *Brassica juncea* (raya/Indian mustard) and *Brassica campestris* var. *toria*/sarson (turnip rape/brown sarson), both part of the Brassicaceae (mustard/cabbage) family. Rapeseed and canola-type varieties, which are lower in erucic acid and glucosinolates, are also increasingly promoted alongside traditional mustard types.

3. Importance in Pakistan

Mustard and rapeseed together form an important part of Pakistan's domestic edible oilseed base, though the country still imports the large majority of its edible oil needs, mainly as palm oil, making local oilseed expansion a national priority. Mustard requires less water and can be grown on land unsuitable for wheat or other high-input rabi crops, making it valuable for farmers in low-rainfall or salt-affected areas. Its oil cake (the residue after oil extraction) is also a useful protein-rich livestock feed, adding to the crop's overall value for mixed farming households.

4. Major Producing Provinces and Districts

- Punjab: Rahim Yar Khan, Bahawalpur, Muzaffargarh, Layyah, and parts of southern Punjab, along with scattered rainfed areas in the Pothohar region.
- Sindh: Sanghar, Mirpurkhas, Umerkot, and Tharparkar — grown both as a sole crop and for fodder/oil combined.
- Khyber Pakhtunkhwa: D.I. Khan, Bannu, and some northern districts, mostly on rainfed or low-input land.
- Balochistan: Limited but present in Nasirabad and Jaffarabad districts where irrigation is available alongside wheat.

5. Climate Requirements

Mustard is a cool-season crop, growing best with temperatures between 10°C and 25°C, and it can tolerate light frost better than many other oilseed crops, though severe frost during flowering can reduce pod set. It requires relatively low rainfall or irrigation compared to wheat and can perform reasonably well on residual soil moisture, making it well-suited to Pakistan's drier rabi tracts. Excess humidity during flowering, however, favors fungal diseases such as *Alternaria* blight.

6. Soil Requirements

Mustard grows well across a wide range of soils, from sandy loam to clay loam, and tolerates mild to moderate soil salinity better than most other oilseed and pulse crops, which is one reason it is often grown on marginal land in Punjab and Sindh. A soil pH between 6.0 and 7.5 is ideal, but the crop can perform reasonably even on slightly alkaline soils common in Pakistan's irrigated plains, provided drainage is adequate.

7. Land Preparation

Land is normally ploughed two to three times followed by planking to prepare a fine, firm seedbed, since mustard seed is small and needs good soil-to-seed contact for uniform germination. Well-rotted farmyard manure, where available, can be incorporated ahead of sowing to improve soil fertility, though mustard is often grown with fewer inputs than wheat on the same farm.

8. Recommended Varieties (Pakistan)

- Punjab Sarson/Raya types and locally recommended varieties released by the Oilseeds Research Institute at AARI Faisalabad and the Punjab Seed Corporation.
- Canola-quality (low erucic acid, low glucosinolate) varieties are being promoted nationally to improve oil quality and meet international edible oil standards, supported by NARC's oilseed program.
- Local Sindh and Sindh Agriculture Research Institute recommended varieties adapted to that province's soils and climate.
- Farmers are advised to check the latest seasonal variety list from provincial seed corporations, since new canola-type releases are gradually replacing older, higher erucic acid mustard varieties.

9. Seed Rate

The recommended seed rate for mustard is about 1.5–2 kg per acre when drilled in rows, slightly higher (2–2.5 kg per acre) if broadcast, since broadcasting typically gives less uniform seed placement and stand establishment.

10. Seed Treatment

Seed should be treated with a recommended fungicide such as thiram or carbendazim (2–3 g per kg seed) to reduce damping-off and seedling blight risk, especially important given mustard's small seed size and shallow sowing depth, which make young seedlings vulnerable to soil-borne pathogens.

11. Sowing Time (Province-wise)

- Punjab: Sown mid-September to mid-October for the main rabi crop; slightly earlier in cooler northern districts.

- Sindh: Sown October–November, adjusted to local temperature patterns.
- Khyber Pakhtunkhwa: Sown late September to October in the plains.
- Balochistan: Sown October, coinciding with wheat sowing where irrigation allows a combined cropping pattern.

12. Sowing Methods

Mustard is generally sown by drilling seed in rows 30–45 cm apart at a shallow depth of about 2–3 cm, since deeper sowing reduces emergence due to the seed's small size and limited food reserves. Broadcasting followed by light ploughing is also practiced, particularly by smallholders growing mustard as a secondary or border crop, but row sowing with a seed drill generally gives better stand uniformity and easier weeding.

13. Fertilizer Recommendations (NPK)

A general recommendation is 60–80 kg Nitrogen, 40–50 kg Phosphorus (P₂O₅), and 25–30 kg Potash (K₂O) per acre, adjusted based on soil test results and expected yield potential. Half of the nitrogen and the full doses of phosphorus and potash are applied at sowing, with the remaining nitrogen top-dressed at the first irrigation or about 30 days after sowing, coinciding with active vegetative growth before flowering.

14. Irrigation Schedule

Mustard is more drought-tolerant than many other rabi crops, and in many parts of Punjab and Sindh it is grown with only two to three irrigations: one at sowing (or relying on residual moisture), one at the pre-flowering/rosette stage, and one during pod development, which is the most water-sensitive stage. Waterlogging should be avoided, since mustard is sensitive to standing water, particularly on heavier soils.

15. Weed Management

Mustard's canopy closes relatively quickly once established, giving it some natural weed suppression, but weeds can still reduce yield significantly if left uncontrolled in the first 30–40 days. One to two hand weeding or hoeings during this early period are generally recommended, along with proper field sanitation to reduce weed seed carryover between seasons.

16. Insect Pests

The mustard aphid is the most significant insect pest in Pakistan, capable of causing serious yield loss by sucking sap from stems, leaves, and developing pods, especially in warm spells during the growing season. Painted bug and sawfly can also damage seedlings and pods in some areas. Regular field monitoring, encouraging natural predators such as ladybird beetles, and timely, need-based insecticide application when aphid populations exceed threshold levels are the standard management approach.

17. Diseases

Alternaria blight (leaf spot) is the most common fungal disease of mustard in Pakistan, causing dark spots on leaves and pods that reduce photosynthesis and seed quality, particularly under humid conditions. White rust and downy mildew can also occur, sometimes together causing distinctive stag-head deformities on flowering shoots. Management includes crop rotation, resistant or tolerant varieties, timely sowing to avoid peak disease-favorable weather, and fungicide sprays when disease pressure is high.

18. Deficiency Symptoms

Nitrogen deficiency shows as overall pale yellowing of older leaves and reduced branching. Sulfur deficiency is particularly important in mustard because sulfur is a key component of the oil and flavor compounds in Brassica crops; deficiency causes pale, yellowish young leaves and reduced oil content and quality. Boron deficiency can cause poor pod set and hollow stem symptoms, a problem sometimes seen on light, sandy soils.

19. Harvesting

Mustard is ready for harvest when pods turn yellowish-brown and seeds inside rattle when shaken, typically 120–150 days after sowing depending on variety and sowing date. Because pods shatter easily when overripe, timely harvesting is important to avoid seed loss; the crop is usually cut by hand or with a reaper, left to dry in the field for a few days, and then threshed, traditionally by beating or trampling, though mechanical threshers are increasingly common on larger farms.

20. Storage

Threshed mustard seed should be dried to a safe moisture content (below about 8–9%) before storage to prevent mold growth and maintain oil quality. Seed is generally stored in clean, dry gunny bags or bins in a cool, well-ventilated store, protected from moisture and storage pests, since damp storage conditions can quickly reduce both germination and oil quality.

21. Expected Yield in Pakistan

The national average yield of mustard and rapeseed in Pakistan is relatively low, typically around 700–900 kg per hectare, reflecting the crop's common status as a low-input, marginal-land crop. With better varieties, adequate fertilization, timely sowing, and proper aphid and disease control, yields of 1,300–1,600 kg per hectare are achievable, showing significant scope for improvement through better management rather than expanded area alone.

22. Government Recommendations

The Oilseeds Research Institute at AARI Faisalabad and the National Agricultural Research Centre run active breeding and promotion programs for canola-quality mustard and rapeseed varieties, aiming to improve both yield and oil quality to help reduce Pakistan's heavy reliance on imported edible oil. Government edible oil policies periodically include support price mechanisms and certified seed subsidies to encourage farmers to expand oilseed cultivation, including mustard, on suitable rabi land.

23. Modern Farming Practices

Line sowing with seed drills instead of broadcasting is being promoted to improve plant stand, ease of weeding, and fertilizer placement efficiency. Timely sowing within the recommended window, balanced fertilizer use guided by soil testing, and adoption of canola-quality varieties are the main modern practices being extended to farmers through provincial agriculture departments and oilseed development programs.

24. Precision Agriculture

Precision practices for mustard remain limited but growing, including soil testing to guide sulfur and boron application (both important for oil quality), and satellite or mobile-based advisories to help farmers time sowing to make the best use of available soil moisture. Some research programs are also exploring drone-based monitoring for aphid outbreaks across larger mustard-growing blocks.

25. Sustainable Farming

Mustard fits well into sustainable, water-efficient cropping systems because of its relatively low irrigation requirement compared to wheat, its tolerance of mild soil salinity, and the feed value of its oil cake by-product for livestock. Promoting crop rotation with cereals, reducing unnecessary pesticide use through integrated pest management for aphids, and encouraging balanced rather than excess nitrogen use are the main sustainability priorities for the crop in Pakistan.

26. Regional Notes

Mustard's tolerance for lower rainfall, moderate salinity, and marginal soils makes it a particularly valuable option for farmers in southern Punjab, Sindh's saline tracts, and parts of Balochistan, where it can be grown when higher-input crops are not feasible. Encouraging wider adoption of improved canola-quality varieties, together with better aphid and disease management, offers Pakistan a realistic path to boosting domestic edible oil production from this traditionally under-invested but resilient crop.